

■■ NetBIOS & SMB Enumeration Lab Report

Kali Linux | Ethical Hacking | Cybersecurity Practice

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Focus:	Penetration Testing & Network Reconnaissance
Tools:	Kali Linux, Nmap, Enum4linux
Environment:	Isolated Virtual Lab

1. Overview

This report demonstrates network reconnaissance and SMB enumeration using NetBIOS services in a controlled Kali Linux lab environment. The objective is to understand information exposure during penetration testing.

2. Objectives

- Understand NetBIOS and SMB communication
- Perform network scanning and enumeration
- Extract system, user, and share information
- Practice ethical hacking in a safe lab environment

3. Tools Used

- Kali Linux
- Nmap
- Enum4linux
- Virtual Lab Environment

4. NetBIOS Summary

NetBIOS is a legacy protocol used for local network communication, including name resolution and session management.

Ports:

- 137 UDP - Name Service
- 138 UDP - Datagram Service
- 139 TCP - Session Service

5. Practical Commands

```
nmap -p 137,138,139 -sV <IP>  
enum4linux -a <IP>  
enum4linux -U <IP>  
enum4linux -S <IP>
```

6. Key Findings

- SMB services detected
- User accounts enumerated
- Shared folders discovered
- System information exposed through enumeration

7. Disclaimer

This report is for educational purposes only. All testing was performed in a controlled lab environment. Unauthorized scanning or access to systems is illegal.